Distributed LinUCB-Based Adaptive DB-LBT with Local Observations for Wi-Fi/NR-U Coexistence

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***Abstract*—Deterministic backoff with listen before talk (DB-LBT) uses local countdown interruptions to reduce repeated Wi-Fi/NR-U collisions, but one recovery profile need not suit both lightly and heavily loaded cells. We formulate its adaptation as a protocol-constrained contextual bandit under locally observable MAC information. Twenty-four legal DB-LBT profiles are screened offline. At deployment, each transmitter runs an independent LinUCB controller over two profiles whose ranking reverses between a $4+4$ light-load regime and a $6+6$ high-load regime. Its 11 features contain only local CCA, queue, delay, retransmission, and ACK/HARQ history; neither peer state nor global evaluation utility enters online learning.**

We state the sensing, reception, local-separability, and timescale conditions needed for such history to remain useful despite random fading and traffic. Across 32 untouched seeds and eight alternating 4096-round phases, adaptive minus fixed-TMC utility is $+0.007338$ under phase-conditional aggregation (95% CI $[+0.007093, +0.007584]$; 32/32 positive). The gain is $+0.015735$ in the light phase and -0.001058 in the high phase. Median and 95th-percentile recovery times are 736 and 928 rounds, both below the phase dwell time. A whole-run mixed-population diagnostic is also reported and is negative, showing why the operating regime and aggregation rule must be explicit.

***Index Terms*—Wi-Fi, NR-U, coexistence, listen before talk, deterministic backoff, contextual bandit, local observation.**

I. Introduction

Wi-Fi and 5G New Radio in unlicensed spectrum (NR-U) contend without a common scheduler. Both defer to detected energy and count down a backoff, yet their surrounding procedures differ. Wi-Fi follows DCF/EDCA after an inter-frame deferral. NR-U category-4 LBT also observes channel-occupancy limits and aligns access to an NR transmission boundary [1]–[3]. Simultaneous counter expiry can therefore waste airtime across both technologies.

DB-LBT replaces part of repeated random recovery with a counter derived from locally observed countdown interruptions [4]. The mechanism can approach a collision-reduced order without node identities or cross-technology messages. Its later TMC form remains effective under non-full-buffer traffic, random busy periods, sensing perturbations, and asymmetric energy detection [5]. Those cases also expose a control problem. A recovery tuple that admits intermittent queues quickly can be too aggressive after more contenders become active, whereas a conservative tuple can add avoidable delay in a light cell.

The instantaneous radio channel is not predictable from a short MAC history. It includes nonlinear CCA threshold crossings, fading, interference, and decoding outcomes. A transmitter nevertheless observes a slower protocol trace: countdown freezes, queue growth, access delay, retransmissions, and ACK/HARQ outcomes. When load or occupancy persists for several update windows, that trace can identify which DB-LBT profile works better without reconstructing the channel itself. This motivates our main formulation: *DB-LBT parameter adaptation is a protocol-constrained contextual bandit under locally observable MAC information.*

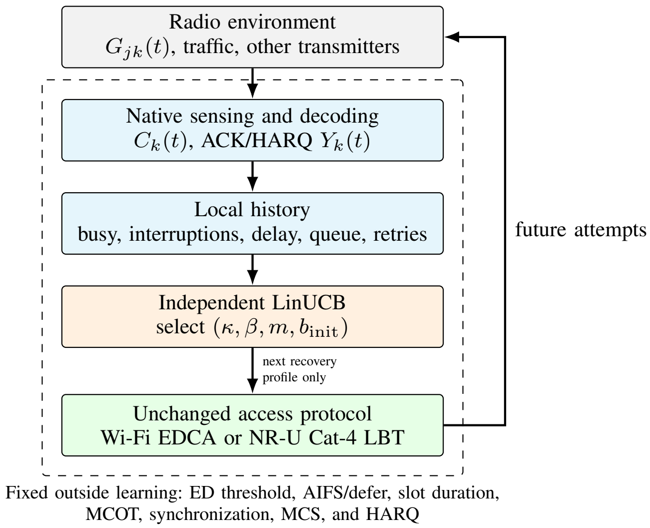
The formulation targets mutually detectable indoor cells with piecewise-stable offered load or active population. The context must be stable long enough to learn, yet different enough across phases to make adaptation worthwhile. It is not intended for per-packet channel prediction.

The contributions are threefold. First, we map DB-LBT recovery control to a contextual bandit whose actions cannot alter CCA thresholds, Wi-Fi deferral, NR-U synchronization or occupancy limits, PHY rate, or retransmission semantics. The 24-profile design space is screened offline and reduced to two conflicting profiles for online selection. Second, we connect the 11 local features to protocol events and state explicit sensing, reception, separability, and dwell-time conditions under which they are informative. Third, a paired 32-seed experiment measures both performance and adaptation time across repeated $4+4$ and $6+6$ phases. It finds a positive phase-conditional effect, identifies a small high-load penalty, and preserves the opposite whole-run diagnostic rather than hiding aggregation sensitivity.

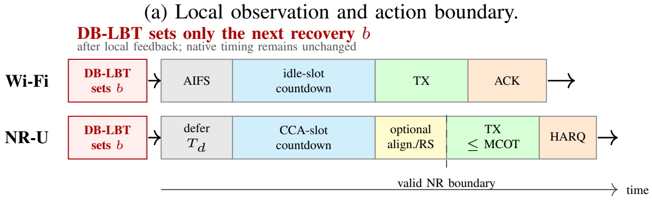
II. Related Work

Wi-Fi/NR-U coexistence schemes tune contention windows, reserve airtime, resolve mini-slot collisions, or schedule around technology-specific timing [3], [6], [7]. Deterministic and semi-random WLAN backoff instead creates post-success structure without explicit slot assignments [8], [9]. DB-LBT extends that idea across Wi-Fi and NR-U: interruptions estimate active contention and set a deterministic recovery counter [4], [5]. We retain this mechanism and adapt only its existing recovery parameters.

Machine learning has been used throughout the IEEE 802.11 stack [10]. LTE-LAA studies adapt contention windows from



(a) Local observation and action boundary.



(b) Insertion point in Wi-Fi and NR-U access.

Fig. 1. Each transmitter selects only its next DB-LBT recovery profile. Native CCA, countdown, synchronization, MCOT, and ACK/HARQ behavior remain outside the learner.

observed performance [11], and reinforcement learning can optimize technology-level efficiency and fairness [12]. Our controller has a narrower action boundary and no shared state. LinUCB supplies uncertainty-aware selection for a small discrete set using short local histories [13]; the contribution is the protocol mapping and its operating conditions, not a new bandit solver.

III. Protocol and Local-Observation Model

A. DB-LBT Insertion Point

Let $\mathcal{K} = \mathcal{K}_{\mathrm{W}} \cup \mathcal{K}_{\mathrm{N}}$ contain the Wi-Fi APs and NR-U gNBs sharing one channel. For local countdown slot ℓ, node k updates

$$c_k(\ell+1) = \begin{cases} c_k(\ell) - 1, & C_k(\ell) = 0, \\ c_k(\ell), & C_k(\ell) = 1, \end{cases} \tag{1}$$

where $C_k = 1$ is a busy CCA sample. A maximal busy episode freezes the counter and increments interruption count i_k. In a mutually detectable collision domain, successful peer transmissions create distinct busy episodes, which motivates the DB-LBT estimate $\hat{n}_k \approx 1 + i_k$.

Figure 1 locates the control boundary. DB-LBT supplies the next recovery counter after an ACK/HARQ outcome. Wi-Fi still applies its inter-frame deferral and EDCA rules; NR-U still applies its priority class, maximum channel occupancy, and transmission-boundary alignment [2], [7].

TABLE I
Local Features and Their Protocol Origin

Feature group	Local source	Action-relevant cue
Busy fraction, interruptions	CCA and countdown freezes	Active peers or persistent occupancy
Success, failure, retries	ACK/HARQ and retry state	Service after attempted access
Queue, arrivals, delay	Local traffic and MAC queue	Activation and congestion
Effective-data occupancy, busy share	Successful channel use	Schedule efficiency and local share

B. What Local History Can Represent

Propagation gain $G_{jk}(t)$ includes path loss, shadowing, and small-scale fading. CCA and reception compress that nonlinear channel into protocol outcomes:

$$C_k(t) = \mathbb{1}\left\{ N_k + \sum_{j \neq k} a_j(t) P_j G_{jk}(t) \geq \Gamma_k \right\}, \tag{2}$$

$$\Pr\{Y_k = 1 \mid \gamma_k, u_k\} = 1 - \mathrm{PER}_{\tau_k}(\gamma_k, u_k). \tag{3}$$

A failed packet can thus result from simultaneous access, fading, hidden interference, or decoding. The learner never treats one failure as a collision measurement.

Three conditions connect these outcomes to the event-level contention model: (A1) every co-channel transmission exceeds every contender's CCA threshold; (A2) isolated reception succeeds with high probability; and (A3) simultaneous transmissions have no capture. Then

$$\begin{aligned} C_k(\ell) &= \mathbb{1}\left\{ \sum_{j \neq k} a_j(\ell) > 0 \right\}, \\ Y_k(\ell) = 1 &\iff \sum_j a_j(\ell) = 1. \end{aligned} \tag{4}$$

Busy episodes now carry contention information: one zero counter succeeds and multiple zeros collide. Equation (4) is an abstraction for the event experiment, not a claim that a fading PHY is linear.

The controller aggregates 11 measurements already available at an AP or gNB: success and failure ratios, CCA busy fraction, countdown interruptions, delay EWMA and P95, queue occupancy, arrivals, retransmissions, effective-data occupancy, and $\hat{n}_k = \mathrm{clip}(1 + \bar{i}_k, 1, 32)$. CCA and interruptions indicate occupancy; queue and delay reveal demand; ACK/HARQ and retransmissions describe service after access. Periodic occupancy or a changed active set need not be explicitly labeled if it leaves a distinct local trace.

Let z denote a MAC occupancy regime, $q^\star(z)$ its best allowed profile, and $\bar{x}_k(z)$ node k's mean context. Action-relevant local separability requires

$$q^\star(z) \neq q^\star(z') \implies \|\bar{x}_k(z) - \bar{x}_k(z')\|_2 \geq \delta_x > 0. \tag{5}$$

TABLE II
PROFILES ALLOWED DURING ONLINE CONTROL

Profile	κ	β	m	b_{init}	Discovery role
q_4	5	2	10	15	Light $4+4$
q_{20}	7	3	6	15	High $6+6$; TMC

The corresponding timescales are

$$T_{\text{phy}} \ll T_{\text{ctx}} \lesssim T_{\text{adapt}} \ll T_{\text{dwell}} < T_{\text{hor}}. \tag{6}$$

Fast radio and arrival randomness can then average into history before a phase changes, while at least one change occurs within the operating horizon. Equations (5)–(6) define the difficult but testable setting in which local adaptation is useful.

IV. Protocol-Constrained Contextual Bandit

A. Legal Profiles and Offline Screening

When a queue becomes active, node k draws $b_k \sim \mathcal{U}[0, b_{\text{init}}]$. After an attempt, DB-LBT sets

$$b_k^+ = \begin{cases} \alpha + i_k, & r_k \bmod \kappa < \beta, \\ \mathcal{U}[0, m], & \text{otherwise}, \end{cases} \tag{7}$$

where r_k is the retransmission count. The first branch converts observed interruptions into spacing; the second separates coincident schedules. We fix $\alpha = 11$ and form 24 legal profiles from $\kappa \in \{5, 7\}$, $\beta \in \{2, 3\}$, $m \in \{4, 6, 10\}$, and $b_{\text{init}} \in \{15, 31\}$, always with $\beta < \kappa$. This grid contains the published TMC profile $(7, 3, 6, 15)$.

Deploying all 24 profiles would spend scarce online samples on actions that do not address the target regime change. Design-space screening is separated from control. Every profile is evaluated offline across 12 topology, load, traffic, interference, and turnover cells using discovery seeds 9103, 9113, and 9127. A pair advances only if its ordering reverses between two phase-compatible cells, each paired-bootstrap lower bound is positive, and the associated Jain-fairness loss is at most 0.01.

The selected cells are Poisson $4+4$ at 0.025 packets/ms and Poisson $6+6$ at 0.045 packets/ms. Table II gives their representative profiles; q_{20} is fixed TMC.

On the discovery seeds, q_4 exceeds q_{20} by 0.019250 utility in the light cell; q_{20} exceeds q_4 by 0.000354 in the high cell. Online choices are restricted to $\mathcal{Q}_{\text{on}} = \{q_4, q_{20}\}$. This protocol gate is part of the method: LinUCB cannot synthesize a backoff or choose an unvalidated tuple.

B. Distributed LinUCB

Each transmitter holds its own controller and 64-attempt history. For $q \in \mathcal{Q}_{\text{on}}$, it maintains $A_q \in \mathbb{R}^{11\times 11}$ and $d_q \in \mathbb{R}^{11}$. The local working model and selection rule are

$$\begin{aligned} \mathbb{E}[R_{k,t} \mid x_{k,t} = x, q_{k,t} = q] &\simeq x^{\mathsf{T}} \theta_q, \qquad (8) \\ \hat{\theta}_q &= A_q^{-1} d_q, \\ p_q(x_k) &= \hat{\theta}_q^{\mathsf{T}} x_k + c\sqrt{x_k^{\mathsf{T}} A_q^{-1} x_k}, \\ q^* &= \arg\max_{q \in \mathcal{Q}_{\text{on}}} p_q(x_k), \qquad (9) \end{aligned}$$

with ridge 1 and $c = 0.5$. A separate model for each tuple permits parameter interactions without claiming linearity in κ, β, m, or b_{init}. The nonlinear CCA and decoding maps stay outside the learner; only their aggregated MAC consequences enter x_k.

The reward is local. Let A_k be effective-airtime fraction, $D_{95,k}$ P95 delay, s_k successful-busy share, and $P_{f,k}$ failure ratio:

$$\begin{aligned} U_A &= \operatorname{clip}(\hat{n}_k A_k, 0, 1), \\ U_D &= 1 - \operatorname{clip}\left(\frac{D_{95,k}}{16\,\text{ms}\,\hat{n}_k}, 0, 1\right), \\ U_S &= 1 - \operatorname{clip}\left(\frac{|s_k - 1/\hat{n}_k|}{1/\hat{n}_k}, 0, 1\right), \\ R_k &= (U_A + U_D + U_S)/3 - 0.25 P_{f,k}. \qquad (10) \end{aligned}$$

After receiving R_k, only the selected model is updated: $A_q \leftarrow A_q + x_k x_k^{\mathsf{T}}$ and $d_q \leftarrow d_q + R_k x_k$. No global fairness, peer queue, peer action, or cross-technology message appears in this update.

Both profile models are warm-started from 73,541 local (x_k, q, R_k) samples generated by the discovery runs. The global evaluation utility is used to screen the two profiles but never enters this fit. Every node receives the same initial model; its later history and updates are independent. After eight observations, a decision occurs every 32 contention rounds and applies only after the current attempt completes, so credit is not assigned to a countdown begun under an earlier profile.

V. Evaluation

A. Paired Phase Experiment

The event simulator implements the contention reduction in (4). A transmission occupies 2 ms; Wi-Fi and NR-U use equal contention windows [15, 63], and NR-U access is aligned to a 250 µs grid. Exogenous arrival and contention streams are paired between Adaptive DB-LBT and fixed TMC. The 32 formal seeds, ranging from 20029 to 20341, were frozen before the pilot and do not overlap the three discovery seeds.

Each 32,768-round run alternates two 4096-round phases four times. The light phase activates four Wi-Fi and four NR-U nodes with Poisson rate 0.025 packets/ms per node. The high phase activates all six nodes of each technology at rate 0.045. Thus each seed contains seven observed transitions and equal dwell time in both regimes.

TABLE III
FORMAL CONFIRMATION DESIGN

Item	Frozen value
Compared policies	Distributed LinUCB over q_4/q_{20}; fixed q_{20} TMC
Formal evidence	32 paired seeds; discovery seeds excluded
Light phase	$4+4$, Poisson 0.025 packets/ms/node
High phase	$6+6$, Poisson 0.045 packets/ms/node
Schedule	Eight 4096-round phases; 32,768 rounds/run
Inference	Seed-paired bootstrap, 10,000 resamples

The reporting utility is separate from the local reward in (10):

$$U_D^{\text{eval}} = 1 - \text{clip}(D_{95}/500\,\text{ms}, 0, 1),$$
$$U_{\text{eval}} = (A_{\text{total}} + U_D^{\text{eval}} + J)/3 - 0.25P_{\text{col}}. \tag{11}$$

It combines effective data airtime, P95 access delay, Jain fairness, and event collision probability. We first recompute all four components within every contiguous phase, average the eight phase utilities within a seed, and form paired bootstrap intervals across seeds. This phase-conditional estimand keeps the active population fixed inside each metric calculation.

B. Performance and Decisions

Figure 2 shows a strong regime split. In the light $4+4$ phase, Adaptive DB-LBT improves utility by 0.015735 (95% CI $[0.015322, 0.016145]$), and all 32 seed differences are positive. In the high $6+6$ phase the effect is -0.001058 ($[-0.001283, -0.000843]$); only one seed is positive. Averaging equal-duration phases gives $+0.007338$ ($[0.007093, 0.007584]$), positive for every seed.

The component differences explain that result. Relative to TMC, the phase-averaged collision probability falls by 0.032288 and Jain fairness rises by 0.000245. Effective airtime falls by 0.002059, while P95 delay increases by 193.3 µs. The utility gain is therefore a collision reduction that outweighs small airtime and delay costs; it is not a uniform improvement of every metric.

Arm choices follow the offline conflict. Nodes choose q_4 for 98.9% of light-phase decisions. In the high phase, the fixed-TMC profile q_{20} rises to 76.9%. Residual exploration of q_4 is consistent with the small high-load penalty. This behavior also gives a protocol interpretation to the light-load gain: queues repeatedly enter contention, and the shorter recovery cycle with $m = 10$ is favored; persistent high load returns most decisions to TMC.

We measure adaptation from the local reward trace after every explicit phase change. Recovery is the first of three consecutive eight-decision windows that reaches 90% of the improvement from the immediate post-change level to the last-quarter phase level. All 224 transitions recover without censoring. Median T_{adapt} is 736 rounds and the 95th percentile is 928 rounds, well below $T_{\text{dwell}} = 4096$. The experiment therefore checks, rather than only assumes, the critical ordering in (6) for this scenario.

TABLE IV
FIVE-SECOND NS-3 SHADOWING DIAGNOSTIC (TMC, SEED 410)

Tech.	Thr. (Mb/s)	Delay (ms)	IP loss	Access overlap
Wi-Fi	8.121	1.793	0.0117	0.3093
NR-U	7.537	38.087	0.0828	0.7726

C. Aggregation Sensitivity and Packet Check

A whole-run calculation pools phases before forming the nonlinear utility. It gives -0.000668 ($[-0.000921, -0.000413]$), with 6/32 positive seeds. The direction changes because the active population, delay distribution, and fairness denominator are mixed before normalization; the corresponding airtime, P95-delay, and fairness differences become -0.013570, $+2.240\,\text{ms}$, and -0.008170. We retain this diagnostic because an application that values one end-to-end mixed-population score should not substitute the phase-conditional conclusion.

The repository also contains an ns-3.35/5G-LENA/NR-U protocol cross-check with Wi-Fi EDCA, NR-U alignment, HARQ, and a frequency-selective channel. That older three-seed campaign used the initial unrestricted 24-profile model, so its mixed technology-level directions are not treated as confirmation of the restricted controller.

Table IV is a measurement-path diagnostic with 3GPP shadowing enabled, not an adaptive-policy comparison. The access quantity is the fraction of transmission starts overlapping another active transmission; IP loss is computed from transmitted and received packets. Their different values confirm that decoding loss and simultaneous access must remain separate.

VI. DISCUSSION

A. Why Contextual LinUCB Fits This Control Point

The learner ranks two protocol profiles, not instantaneous channel states. Native CCA and ACK/HARQ processing first convert received power and SINR into busy periods, interruptions, outcomes, queues, and delays. A separate linear model per profile then approximates the local reward difference over that 11-dimensional history. This is a deliberately small model for limited online samples. The measured action reversal and $T_{\text{adapt}} < T_{\text{dwell}}$ matter more here than waveform-prediction capacity. If the context distributions overlap or drift inside one update window, a nonlinear learner would not repair the missing information.

The light/high asymmetry is also operationally useful. Adaptation is valuable when queue activation creates a locally visible regime in which TMC is not the best fixed profile. Under persistent high load, TMC is already strong and exploration imposes a small cost. The result therefore supports a conditional deployment rule, not the claim that learning dominates fixed DB-LBT at every load.

B. Restricted Bandit-Game Interpretation

With several adaptive nodes, node k receives $R_k(\boldsymbol{q}_t)$ under joint action $\boldsymbol{q}_t = (q_{1,t}, \ldots, q_{|\mathcal{K}|,t})$. This can be interpreted

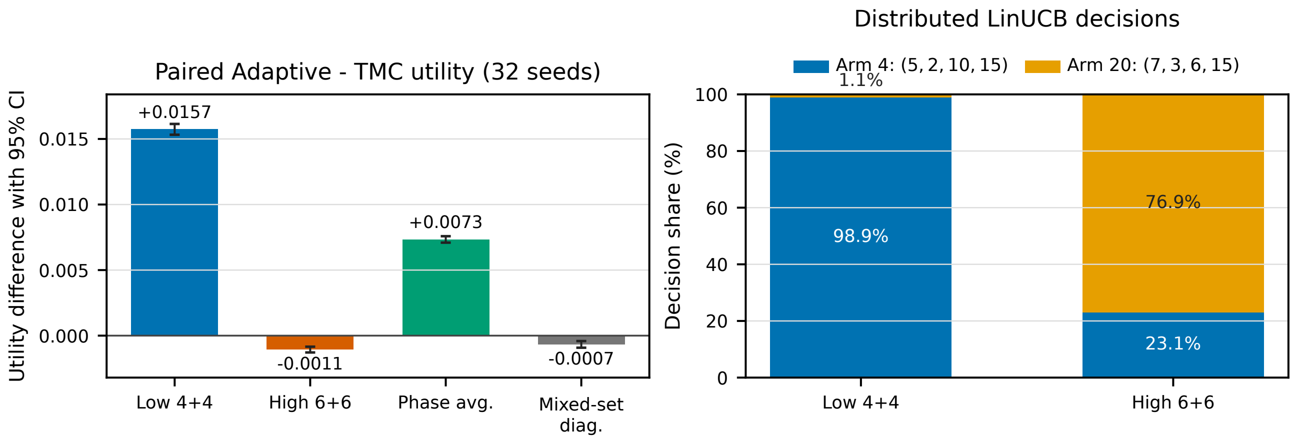


Fig. 2. Independent 32-seed confirmation. Left: Adaptive-minus-TMC utility; intervals are paired 95% bootstrap CIs. "Phase avg." is the primary within-phase estimator, while "Mixed-set diag." computes one utility over the entire run. Right: online choices are restricted to the two profiles in Table II.

as a restricted stochastic contextual bandit game when peer actions are non-anticipating within a decision interval, aggregate peer behavior varies slowly over T_{adapt}, and updates are asynchronous or weakly coupled. Those restrictions let each node treat the local reward process as temporarily learnable. Synchronized strategic adaptation, action-conditioned retaliation, or jamming falls outside this interpretation; the experiment studies empirical recovery, not equilibrium or adversarial regret.

C. Limitations and Next Experiments

The positive result is bounded by A1–A3, two locally separable Poisson regimes, two screened profiles, and the event-level collision abstraction. The whole-run sign reversal further shows that phase utility is appropriate only when regime-conditional service is the target. An end-to-end application with changing membership may prefer a different objective or controller.

Packet-level validation should next port the restricted candidate gate and its warm start into ns-3, then vary log-normal shadowing, fading coherence, CCA margin, and hidden-node geometry. Measurements must keep PHY decoding loss, CCA miss/false-busy events, and simultaneous-access collisions separate. Longer runs should sweep T_{dwell} around the observed 928-round recovery tail. These experiments would reveal whether a sliding window, change-point reset, or nonlinear reward model is warranted.

VII. Conclusion

We cast DB-LBT recovery adaptation as a protocol-constrained contextual bandit driven by local MAC history. Offline screening reduces 24 legal tuples to two profiles with opposite light/high-load rankings; distributed LinUCB then selects between them without shared state or changes to native Wi-Fi and NR-U procedures. In the tested phase-conditional setting, the method improves utility for all 32 formal seeds and recovers well before the next regime change. Its high-load penalty and negative whole-run diagnostic define the boundary of that conclusion as clearly as the light-load gain.

Appendix: Reproducibility Package

The complete source code, frozen configurations, seeds, compact results, figures, and reproduction commands are available at:

https://github.com/Ignister01/distributed-linucb-dblbt

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